

PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student at PES University with a Minor in Computer Science. Experienced in robotics hardware development, mechanical design, electrical schematic design, embedded systems, and HMI development. Hands-on exposure to CAD modeling, autonomous robotics, UAV systems, computer vision, and AI-enabled applications. Passionate about developing integrated robotic systems combining hardware, designing, embedded computing, and intelligent automation.

EXPERIENCE

Research Intern – L&T

Jan 2026 – Present

- Contributed to the hardware development of an autonomous industrial robot, including mechanical design, subsystem integration, and full platform assembly and assisted in developing the BOM.
- Designed the CAD, electrical schematics and wiring architectures for robotic subsystems, ensuring reliable communication and power distribution.
- Developed embedded HMI UI/UX interfaces to improve operator interaction, system monitoring, and usability.
- Participated in technical problem solving, prototype testing, component selection, Gazebo simulation, Path planning and design reviews throughout the product development lifecycle.

AI / Agent Workflow Automation Project

Jun 2025 – Aug 2025

- Developed an AI Agent workflow automation pipeline capable of executing multi-step tasks with minimal human intervention.
- Trained and evaluated machine learning models for task classification, achieving approximately 85% accuracy on evaluation datasets.

UAV Systems Research – IISc UAV Laboratory

Short-Term Research

- Studied UAV flight dynamics, propulsion systems, avionics, and onboard electronics for autonomous aerial platforms.
- Assisted in the design and prototyping of a UAV platform, supporting airframe development, subsystem integration, and hardware selection activities.

PROJECTS

UAV-Based Computer Vision for Pipe Crack Detection

- Developed a computer vision system for detecting cracks in buried pipes using imagery captured by a UAV-mounted camera.
- Built a custom image dataset of pipe interiors and surface defects for model training and evaluation.
- Implemented machine learning techniques for crack detection and automated inspection workflows.

TECHNICAL SKILLS

Mechanical Design: SolidWorks, Fusion 360, GD&T

Robotics & Embedded: CAD designing, ROS, Gazebo, LiDAR, SLAM, Sensor Fusion, NVIDIA Boards, Embedded Systems, prototyping, HMI UI/UX interfaces, NAV2.

Programming: Python, C, C++, JavaScript, SQL, React.js, Node.js, Flask, OOP

AI / ML: Prompt Engineering, Machine Learning, Computer Vision, Model Training, Model Evaluation, LLM's.

Tools & Platforms: Git, VS Code, Cadence, Power BI, Unreal Engine, Godot, Linux

EDUCATION

PES University, Bengaluru

2022 – 2026

B.Tech in Electronics & Communication Engineering

Minor in Computer Science

NRI Junior College, Guntur

2020 – 2022

Andhra Pradesh Board of Education

CORE COMPETENCIES

Autonomous Robotics • Embedded Systems • Sensor Integration • Mechanical Designing • DSA • DBMS • Debugging & Testing • Version Control • Product Development • Technical breadth (Electronics + CS + Robotics)